

ComplyAI

AI-Powered Compliance Gap Analysis Assistant

Turning multi-week manual compliance reviews into a multi-minute AI-assisted analysis — built for a 12-hour hackathon, designed to demonstrate real business value.

Document: Product Requirements Document (Condensed)
Prepared for: Hackathon Judges
Version: 1.0
Status: MVP Scope — 12-Hour Build

1. Executive Summary

ComplyAI is an AI-powered assistant that automatically compares a company's internal policy documents against government regulations, producing a structured compliance gap report in minutes instead of weeks. Users upload two PDFs — a policy and a regulation — and an LLM-driven engine returns a compliance score, risk level, missing requirements, outdated policies, and actionable recommendations.

Why it matters: Compliance review is currently manual, slow, and expensive. ComplyAI replaces days of legal cross-referencing with a single AI-generated report — without replacing legal judgment, only accelerating it.

Aspect	Summary
Core Function	AI-driven gap analysis between company policy and regulation documents
Primary Users	Compliance Officers, Legal Teams, HR, Auditors, Founders
Build Scope	Functional MVP completed within a 12-hour hackathon
Key Differentiator	Instant, structured, actionable output — not a raw document diff

2. Problem Statement

Companies expanding into new states or countries must verify that internal policies meet local regulatory requirements. Today, this relies on manual document review by legal and compliance teams using spreadsheets and side-by-side reading — a process that is slow, costly, and prone to human error.

Current Step	Pain Point
Document intake	No structured or centralized process
Manual comparison	Time-consuming, tedious, hard to scale
Gap tracking (spreadsheets)	Error-prone, inconsistent formatting
Risk assessment	Subjective, varies reviewer to reviewer
Repeat per jurisdiction	Fully manual each time — doesn't scale with growth

3. Proposed Solution

A web application where users upload a company policy PDF and a target regulation PDF. The backend extracts text from both, sends an engineered prompt to an LLM (Gemini 2.5 Flash), and returns a structured JSON compliance analysis — rendered as an interactive dashboard and downloadable report.



4. Product Vision & Goals

Vision: Become the fastest way for any company to understand its regulatory exposure in a new market — reducing a multi-week legal review to a multi-minute AI-assisted analysis.

Goal	Hackathon Target
Speed	Full analysis in under 60 seconds
Clarity	Non-lawyers can understand the output unaided
Credibility	Output feels defensible and demo-ready for judges
Simplicity	No login, no setup — upload and go

5. Target Users & Personas

Segment	Need
Compliance Officers	Fast first-pass gap identification
Legal Teams	Structured triage before deep manual review
HR Teams	Verify policy alignment with local labor law
Internal Auditors	Documented evidence for audit trails
Startup Founders	Affordable compliance check without in-house legal
Enterprise Companies	Scalable review across many jurisdictions

Persona — Priya, Compliance Officer

Mid-size SaaS company. Needs a fast way to check HR policy against new state labor law. Wants clarity, not legal jargon.

Persona — Aisha, Startup Founder

Expanding into a new country with no in-house legal team. Wants a simple, affordable first-pass compliance check before hiring outside counsel.

6. Key Features

Feature	Priority	Description
Dual PDF Upload	MVP	Upload company policy + regulation document
AI Gap Analysis	MVP	LLM compares documents, returns structured JSON
Compliance Dashboard	MVP	Score, risk level, gaps, recommendations visualized
Report Download	MVP	Export findings as a shareable report
Multi-Regulation Compare	Nice to Have	Compare one policy across multiple jurisdictions
Regulation Watch	Future	Auto-detect regulation changes over time

7. Functional Requirements

Category	Requirements
Must Have (MVP)	PDF upload (policy + regulation), text extraction, AI gap analysis, JSON output, dashboard, report download, error handling, loading state
Nice to Have	Clause-level source highlighting, multi-regulation comparison, shareable report link
Future Scope	Regulation watch, auto regulation updates, version history, trend charts, multi-language support

8. Non-Functional Requirements

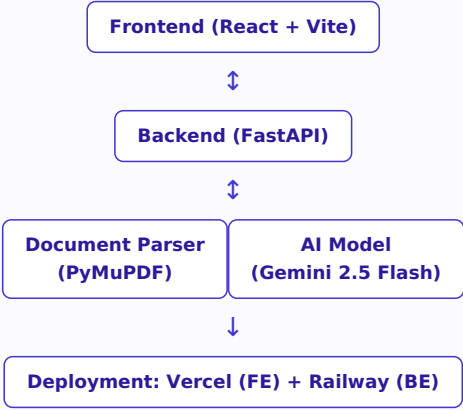
Category	Requirement
Performance	Analysis completes in under 60 seconds for typical documents
Security	No persistent storage of sensitive docs beyond session; server-side API keys
Usability	Zero-training-needed upload → result flow
Scalability	Stateless backend, scalable post-hackathon
Accessibility	High-contrast dark theme, readable typography
Reliability	Graceful error states for AI, parsing, and network failures

9. User Journey / User Flow

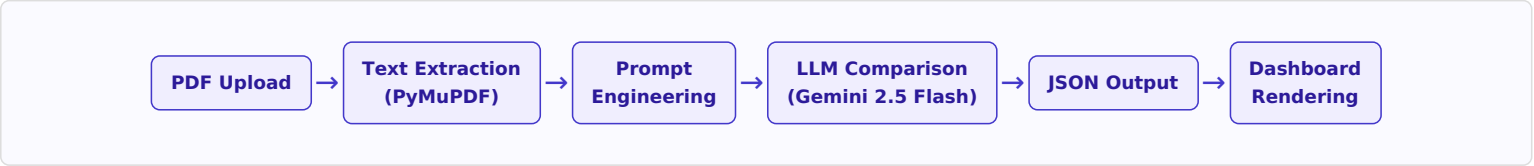


Judge experience: the entire flow above — from landing page to a downloaded report — is designed to be demonstrable in under 60 seconds.

10. System Architecture



11. AI Workflow



The system prompt instructs the model to act as a compliance analyst and return strict JSON matching a fixed schema (score, risk level, summary, missing requirements, outdated policies, recommendations). A low temperature (0.2–0.3) keeps output consistent; a single automatic retry handles malformed JSON.

12. Tech Stack

Layer	Technology
Frontend	React + Vite
Backend	FastAPI
AI Model	Gemini 2.5 Flash
PDF Parsing	PyMuPDF
HTTP Client	Axios
Deployment	Vercel (frontend) + Railway (backend)
Version Control	Git + GitHub

13. Success Metrics

Metric	Target
End-to-end flow reliability	Works without failure during live demo
Analysis speed	Under 60 seconds per document pair
Output coherence	Score and gaps are defensible when explained to judges
Demo stability	Zero crashes during presentation

14. Future Scope

- Regulation Watch — continuous monitoring for regulatory changes
- Automatic regulation updates from official government sources
- Version history to track compliance improvement over time
- Compliance trend dashboards across multiple jurisdictions
- Multi-language document support
- Multi-regulation comparison in a single analysis

15. Timeline / Roadmap — 12-Hour Hackathon Plan

Time	Activity
Hr 0-1	Finalize scope, repo setup, environments, API keys
Hr 1-3	Backend + frontend skeletons (routes, upload pages)
Hr 3-5	AI integration: prompt engineering, Gemini call, JSON validation
Hr 5-8	End-to-end connection + compliance dashboard UI build
Hr 8-10	Error handling, loading states, report download, UI polish
Hr 10-11.5	Testing, bug fixes, deployment (Vercel + Railway)
Hr 11.5-12	Demo rehearsal and presentation prep

16. Risks & Assumptions

Risk	Mitigation
AI returns malformed JSON	Server-side validation + one automatic retry
API rate limits hit during demo	Cache a pre-run example result as fallback
Large PDFs slow extraction	Enforce file size limits, pre-test with real documents
Time overrun	Strict MVP scope; cut Nice-to-Have features first

Key Assumptions: Uploaded PDFs are text-based (not scanned images), documents are in English, and a single analysis session is sufficient for the MVP — no cross-session persistence required.

17. Conclusion

ComplyAI demonstrates how a focused AI application can turn a slow, manual, expensive compliance process into an instant, structured, and actionable report. The MVP is scoped deliberately — no authentication, no payments, no admin complexity — so the team can ship a working, demo-ready product within 12 hours while still showing clear, credible business value to hackathon judges.